

MARKS: 30

TIME: 1.5 Hrs.

Instruction: (i) Assume necessary data wherever required.
(ii) The figure to the right indicates marks.

Q-1 Attempt the Following

- (a) (i) Draw the spectrum of Fig.1 and Fig.2.
(ii) Draw the spectrum that results from the convolution of two images shown in Fig.1 and Fig.2.

Fig.1: a horizontal sine wave of frequency 8

Fig.2: a vertical sine wave of frequency 16

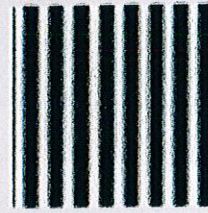


Fig.1



Fig.2

- (b) Find $F(u,v)$ for the following and also draw the spectrum of it.

(i) $f(x,y) = \delta(x,y)$

(ii) $f(x,y) = 0.5(\delta(x,y-a) + \delta(x,y+a))$

Q-2 Attempt Any Three

- (a) (i) Normally, histogram equalization is a global process, explain how it can be used for local enhancement.

(ii) When performing local histogram equalization, why use the sliding window method?

(iii) Why use the block method?

- (b) What is the primary difference in the results from an arithmetic mean compared to a Gaussian spatial filter? Generate the 5×5 gaussian filter mask using PASCAL's triangle. (03)

- (c) Using the concept of image reconstruction using n-bit planes, find the new values of pixel by combining the 8th bit plane and 7th bit plane for the bold pixels (167 & 111) of given image A of Fig.3. (03)

$$A = \begin{bmatrix} \mathbf{167} & 133 & \mathbf{111} \\ 144 & 140 & 135 \\ 159 & 154 & 148 \end{bmatrix}$$

Fig.3

- (d) Consider the 2-bit image of size 5×5 as shown in Fig.4: Calculate the mean using the two formula as discussed in the class and compare the answer using both formulas. (03)

0	0	1	1	2
1	2	3	0	1
3	3	2	2	0
2	3	1	0	0
1	1	3	2	2

Fig.4

Q-3 Attempt The Following

- (a) Consider the image segment shown in Fig.5. (04)

(i) Let $V = \{0, 2, 3\}$, Find the pixel relationship for $N_4(p)$, $N_D(p)$, and $N_8(p)$. (Write only coordinates) CO2(ii) Let $V = \{0, 3\}$, Compute the lengths of the shortest 4-, 8-, and m -path between p and q . If a particular path does not exist between these two points, explain why?

0 (0,0)	1	1	2	3 (q)
1	1	2	0	1
2	2	3	3	2
1	3	0	0 (p)	1
0	0	2	0	1

Fig.5

- (b) Fig.6 shows a horizontal gray-level profile (scan line) of the image that contains various solid objects and line. The number of gray levels is simplified to only seven levels. The profile is a one-dimensional function. Find the first and second derivatives of the image strip. **Also, write comments on first and second derivatives results.** (To avoid a situation in which the previous or next points are outside the range of the scan line, show derivative computations from the second through the penultimate points in the sequence.)

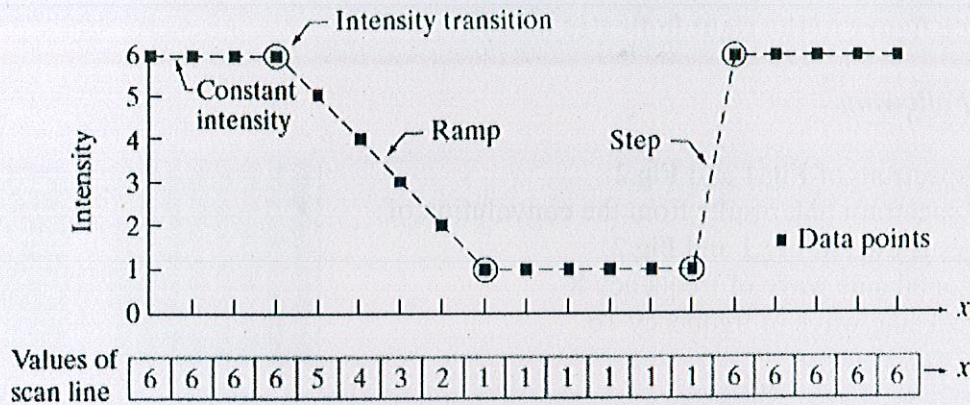


Fig.6

Q-4 Attempt The Following

- (a) Suppose that a 3-bit image ($L=8$) has the intensity distribution as Table I. It is desired to transform the original histogram to a specified histogram as shown in Table II. What will be the actual histogram after the histogram matching operation? Find the transformation between r and z . **Note: Do not round histogram equalized values to nearest integer until final mapping table is avoided ambiguities.**

Also, draw the graph for (i) $P_z(z_q) \text{ v.s. } z_q$ (ii) $P_s(s_k) \text{ v.s. } s_k$

TABLE I

r_k	0	1	2	3	4	5	6	7
n_k	10	5	5	11	5	0	3	6

TABLE II

z_q	0	1	2	3	4	5	6	7
$P_z(z_q)$	5	5	5	10	10	5	5	0

- (b) Suppose an image has a probability density function given in Fig.7(a). We would like to modify it so that it has a probability density function given in Fig.7(b). Derive the transformation function $g(f)$ that will accomplish this. For simplicity, assume both the original image and the modified image can take on gray levels in the continuous range of (0,255). Also find the value of derived function $(g(f))$ at $f=0$ and $f=255$.

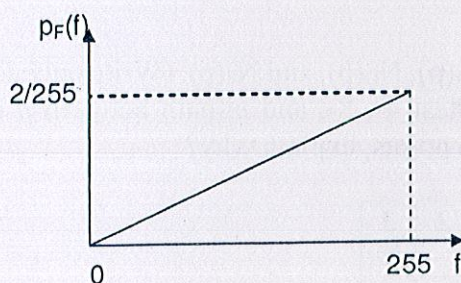


Fig.7(a)

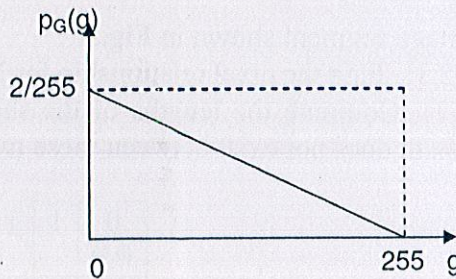


Fig.7(b)

BEST OF LUCK